

DTSA 5799

# Unsupervised Text Classification for Marketing Analytics

|                                    |                                        |               |
|------------------------------------|----------------------------------------|---------------|
| FORMAT                             | CATEGORY                               | CREDITS       |
| Online (DTSA)                      | Elective                               | 1             |
| SECTION                            | BEST ROLE MATCH                        | AVG ALIGNMENT |
| Section 4 – Elective               | Machine Learning Engineer (64.9 / 100) | 56.0 / 100    |
| SPECIALIZATION Marketing Analytics |                                        |               |

56

CURRENT AVG SCORE  
across 6 job roles

94

POTENTIAL SCORE  
with recommended additions

+52

MAX SCORE GAIN  
best single-role improvement

## CURRENT COURSE DESCRIPTION

Marketing data is often so big that humans cannot read or analyze a representative sample of it to understand what insights might lie within. In this course students will learn how to use unsupervised deep learning to train algorithms to extract topics and insights from text data. Students will walk through a conceptual overview of unsupervised machine learning, and dive into real-world datasets through instructor-led tutorials in Python. The course will conclude with a major project.

## INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE              | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|------------------------------|------------|----------|
| Machine Learning Engineer | <div><div></div></div> 64.9 | <div><div></div></div> 98.0  | +33.0      | HIGH     |
| ML Engineer               | <div><div></div></div> 62.9 | <div><div></div></div> 92.8  | +29.9      | MEDIUM   |
| Data Scientist            | <div><div></div></div> 61.0 | <div><div></div></div> 100.0 | +39.0      | HIGH     |
| AI Engineer               | <div><div></div></div> 60.7 | <div><div></div></div> 95.2  | +34.5      | HIGH     |
| Applied Scientist         | <div><div></div></div> 47.5 | <div><div></div></div> 99.4  | +51.9      | HIGH     |
| MLOps Engineer            | <div><div></div></div> 39.1 | <div><div></div></div> 77.5  | +38.4      | HIGH     |

## RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**System Design** 3 of 6 roles

systems

**LLMs & Generative AI** 2 of 6 roles

storage

**Deep Learning** 1 of 6 roles

training

**Business Communication** 1 of 6 roles

business

## PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

4 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

## GAP KEYWORDS BY ROLE

**Machine Learning Engineer**

64.9 → 98.0 (+33.0 pts)

analytics tools business mobile design science models applications

**ML Engineer**

62.9 → 92.8 (+29.9 pts)

discovery systems models world build problems impact product

**Data Scientist**

61.0 → 100.0 (+39.0 pts)

product analytics mitigation social build problems approaches

**AI Engineer**

60.7 → 95.2 (+34.5 pts)

models build technical systems problems technology storage detection

**Applied Scientist**

47.5 → 99.4 (+51.9 pts)

discovery scientific research storage literature problems models technical

**MLOps Engineer**

39.1 → 77.5 (+38.4 pts)

detection systems image training tools models

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.